

Chemical Kinetics

Chapter 3 · All questions with full solutions



Part 1 · Examples

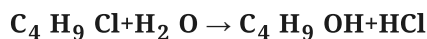
Q

3.1

Average and instantaneous rate from concentration data



From the concentrations of $\text{C}_4\text{H}_9\text{Cl}$ (butyl chloride) at different times given below, calculate the average rate of the reaction:



during different intervals of time.

t / s	0	50	100	150	200	300	400	700	800
$[\text{C}_4\text{H}_9\text{Cl}] / \text{mol L}^{-1}$	0.100	0.0905	0.0820	0.0741	0.0671	0.0549	0.0439	0.0210	0.017

Given

We find the difference in concentration over different intervals of time. Dividing $\Delta[\text{R}]$ by Δt gives the average rate (Table 3.1).

$[\text{C}_4\text{H}_9\text{Cl}]_{t_1} / \text{mol L}^{-1}$	$[\text{C}_4\text{H}_9\text{Cl}]_{t_2} / \text{mol L}^{-1}$	t_1 / s	t_2 / s	$r_{\text{av}} \times 10^4 / \text{mol L}^{-1} \text{s}^{-1}$ $= -\{[\text{C}_4\text{H}_9\text{Cl}]_{t_2} - [\text{C}_4\text{H}_9\text{Cl}]_{t_1}\} / (t_2 - t_1) \times 10^4$
0.100	0.0905	0	50	1.90
0.0905	0.0820	50	100	1.70
0.0820	0.0741	100	150	1.58
0.0741	0.0671	150	200	1.40
0.0671	0.0549	200	300	1.22
0.0549	0.0439	300	400	1.10
0.0439	0.0335	400	500	1.04
0.0210	0.017	700	800	0.4

Substitute

Table 3.1 shows that the average rate falls from $1.90 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$ to $0.4 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. But average rate cannot tell us the rate at one particular instant, since it stays constant over whichever time interval we use to calculate it. So, to get the rate at a particular moment, we find the instantaneous rate. We get this by taking the average rate over the smallest possible time interval, say dt (that is, when Δt approaches zero). So, for an infinitesimally small dt , the instantaneous rate is given by

Key formula

$$r_{\text{av}} = \frac{-\Delta[\text{R}]}{\Delta t} = \frac{\Delta[\text{P}]}{\Delta t} \quad (3.3)$$



Substitute

$$\text{As } \Delta t \rightarrow 0 \text{ or } r_{\text{inst}} = \frac{-d[\text{R}]}{dt} = \frac{d[\text{P}]}{dt}$$

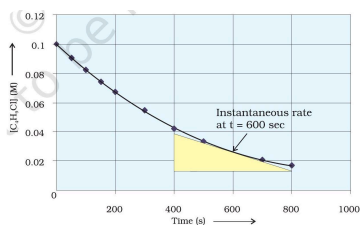


Fig 3.2 Instantaneous rate of hydrolysis of butyl chloride (C_4H_9Cl)

Conclusion

We can also find it graphically. Draw a tangent at time t on either curve (concentration of R or P vs time t) and calculate its slope (Fig. 3.1). So in problem 3.1, we can find r_{inst} at 600s, for example, by plotting the concentration of butyl chloride against time. We draw a tangent that touches the curve at $t = 600$ s (Fig. 3.2).

The slope of this tangent gives the instantaneous rate.

$$\text{So, } r_{\text{inst}} \text{ at } 600 \text{ s} = - \left(\frac{0.0165 - 0.037}{(800 - 400) \text{ s}} \right) \text{ mol L}^{-1} \text{ s}^{-1} = 5.12 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$\text{At } t = 250 \text{ s } r_{\text{inst}} = 1.22 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$t = 350 \text{ s } r_{\text{inst}} = 1.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$t = 450 \text{ s } r_{\text{inst}} = 6.4 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$$

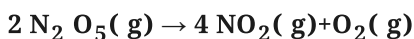
Q

3.2

Average rate of N_2O_5 decomposition in different time units



The decomposition of N_2O_5 in CCl_4 at 318K has been studied by monitoring the concentration of N_2O_5 in the solution. Initially the concentration of N_2O_5 is 2.33 mol L^{-1} and after 184 minutes, it is reduced to 2.08 mol L^{-1} . The reaction takes place according to the equation



Calculate the average rate of this reaction in terms of hours, minutes and seconds. What is the rate of production of NO_2 during this period?

Substitute

$$\begin{aligned} \text{Average Rate} &= \frac{1}{2} \times \frac{\Delta[N_2O_5]}{\Delta t} = - \frac{1}{2} \times \frac{(2.08 - 2.33) \text{ mol L}^{-1}}{184 \text{ min}} \\ &= 6.79 \times 10^{-4} \text{ mol L}^{-1} / \text{min} = (6.79 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}) \times (60 \text{ min} / 1 \text{ h}) \\ &= 4.07 \times 10^{-2} \text{ mol L}^{-1} / \text{h} \\ &= 6.79 \times 10^{-4} \text{ mol L}^{-1} \times 1 \text{ min} / 60 \text{ s} \\ &= 1.13 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1} \end{aligned}$$

Conclusion

Remember that

$$\begin{aligned} \text{Rate} &= \frac{1}{4} \times \frac{\Delta[NO_2]}{\Delta t} \\ \frac{\Delta[NO_2]}{\Delta t} &= 6.79 \times 10^{-4} \times 4 \text{ mol L}^{-1} \text{ min}^{-1} = 2.72 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1} \end{aligned}$$

$$\begin{aligned} \text{Average rate} &= 4.07 \times 10^{-2} \text{ mol L}^{-1} \text{ h}^{-1} = 6.79 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1} \\ &= 1.13 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}; \text{ rate of production of } NO_2 = 2.72 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1} \end{aligned}$$

Q

3.3

Overall order from a fractional rate expression

()

Calculate the overall order of a reaction which has the rate expression

Part (a)

$$\text{Rate} = k[A]^{1/2}[B]^{3/2}$$

Conclusion

$$\text{Rate} = k[A]^x[B]^y$$

$$\text{order} = x+y$$

$$\text{So order} = 1/2 + 3/2 = 2, \text{ i.e., second order}$$

Second order

///

Part (b)

$$\text{Rate} = k[A]^{3/2}[B]^{-1}$$

Conclusion

$$\text{order} = 3/2 + (-1) = 1/2, \text{ i.e., half order.}$$

Half order

///

Q

3.4

Reaction order from rate constant units

✧

Identify the reaction order from each of the following rate constants.

Part (i)

$$k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$$

Conclusion

The unit of a second order rate constant is $\text{L mol}^{-1} \text{ s}^{-1}$. So $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ represents a second order reaction.

Second order reaction

///

Part (ii)

$$k = 3 \times 10^{-4} \text{ s}^{-1}$$

Conclusion

The unit of a first order rate constant is s^{-1} . So $k = 3 \times 10^{-4} \text{ s}^{-1}$ represents a first order reaction.

First order reaction

///

Q

3.5

Rate constant of first order N_2O_5 decomposition

✧

The initial concentration of N_2O_5 in the following first order reaction $\text{N}_2\text{O}_5(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) + 1/2\text{O}_2(\text{g})$ was $1.24 \times 10^{-2} \text{ mol L}^{-1}$ at 318 K. The concentration of N_2O_5 after 60 minutes was $0.20 \times 10^{-2} \text{ mol L}^{-1}$. Calculate the rate constant of the reaction at 318 K.

Substitute

For a first order reaction

$$\log \frac{[R]_1}{[R]_2} = \frac{k(t_2 - t_1)}{2.303}$$

$$k = \frac{2.303}{(t_2 - t_1)} \log \frac{[R]_1}{[R]_2}$$

$$= \frac{2.303}{(60 \text{ min} - 0 \text{ min})} \log \frac{1.24 \times 10^{-2} \text{ mol L}^{-1}}{0.20 \times 10^{-2} \text{ mol L}^{-1}}$$

$$= \frac{2.303}{60} \log 6.2 \text{ min}^{-1}$$

$$k = 0.0304 \text{ min}^{-1}$$

k = 0.0304 min⁻¹

///

Q

3.6

Rate constant
from total
pressure
data

The following data were obtained during the first order thermal decomposition of $\text{N}_2\text{O}_5(\text{g})$ at constant volume:



S.No.	Time / s	Total Pressure / (atm)
1.	0	0.5
2.	100	0.512

Calculate the rate constant.

Given

Let the pressure of $\text{N}_2\text{O}_5(\text{g})$ decrease by $2x$ atm. Two moles of N_2O_5 decompose to give two moles of $\text{N}_2\text{O}_4(\text{g})$ and one mole of $\text{O}_2(\text{g})$. So the pressure of $\text{N}_2\text{O}_4(\text{g})$ increases by $2x$ atm, and the pressure of $\text{O}_2(\text{g})$ increases by x atm.

	$2 \text{N}_2\text{O}_5(\text{g})$	$\rightarrow$	$2 \text{N}_2\text{O}_4(\text{g})$	+	$\text{O}_2(\text{g})$
Start $t = 0$	0.5 atm		0 atm		0 atm
At time t	$(0.5 - 2x)$ atm		$2x$ atm		x atm

Key formula

$$p_t = p_{\text{N}_2\text{O}_5} + p_{\text{N}_2\text{O}_4} + p_{\text{O}_2}$$



$$= (0.5 - 2x) + 2x + x = 0.5 + x$$

$$x = p_t - 0.5$$

$$p_{\text{N}_2\text{O}_5} = 0.5 - 2x$$

$$= 0.5 - 2(p_t - 0.5) = 1.5 - 2p_t$$

Substitute

$$\text{At } t = 100 \text{ s ; } p_t = 0.512 \text{ atm}$$

$$p_{\text{N}_2\text{O}_5} = 1.5 - 2 \times 0.512 = 0.476 \text{ atm}$$

Using equation (3.16)

$$k = \frac{2.303}{t} \log \frac{p_i}{p_A} = \frac{2.303}{100 \text{ s}} \log \frac{0.5 \text{ atm}}{0.476 \text{ atm}}$$

$$= \frac{2.303}{100 \text{ s}} \times 0.0216 = 4.98 \times 10^{-4} \text{ s}^{-1}$$

$$k = 4.98 \times 10^{-4} \text{ s}^{-1} //$$

Q

3.7

Half-life of a
first order
reaction

(())

A first order reaction is found to have a rate constant, $k = 5.5 \times 10^{-14} \text{ s}^{-1}$. Find the half-life of the reaction.

Substitute

Half-life for a first order reaction is

$$t_{1/2} = \frac{0.693}{k}$$

$$t_{1/2} = \frac{0.693}{5.5 \times 10^{-14} \text{ s}^{-1}} = 1.26 \times 10^{13} \text{ s}$$

$$t_{1/2} = 1.26 \times 10^{13} \text{ s} //$$

Q

3.8

Proof: 99.9% completion time versus half-life



Show that in a first order reaction, time required for completion of 99.9% is 10 times of half-life ($t_{1/2}$) of the reaction.

Given

When the reaction is completed, 99.9 %, $[R]_n = [R]_0 - 0.999[R]_0$

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

$$= \frac{2.303}{t} \log \frac{[R]_0}{[R]_0 - 0.999[R]_0} = \frac{2.303}{t} \log 10^3$$

$$t = 6.909 / k$$

Conclusion

For half-life of the reaction

$$t_{1/2} = 0.693 / k$$

$$\frac{t}{t_{1/2}} = \frac{6.909}{k} \times \frac{k}{0.693} = 10$$

Q

3.9

Activation energy and Arrhenius factor from two temperatures



The rate constants of a reaction at 500K and 700K are 0.02 s^{-1} and 0.07 s^{-1} respectively. Calculate the values of E_a and A.

Substitute

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$\log \frac{0.07}{0.02} = \left(\frac{E_a}{2.303 \times 8.314 \text{ JK}^{-1} \text{ mol}^{-1}} \right) \left[\frac{700 - 500}{700 \times 500} \right]$$

$$0.544 = E_a \times 5.714 \times 10^{-4} / 19.15$$

$$E_a = 0.544 \times 19.15 / 5.714 \times 10^{-4} = 18230.8 \text{ J}$$

$$k = A e^{-E_a / RT}$$

$$0.02 = A e^{-18230.8 / 8.314 \times 500}$$

$$A = 0.02 / 0.012 = 1.61$$

$$E_a = 18230.8 \text{ J}; A = 1.61 //$$

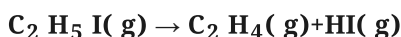
Q

3.10

Rate constant of ethyl iodide decomposition at a new temperature



The first order rate constant for the decomposition of ethyl iodide by the reaction



at 600K is $1.60 \times 10^{-5} \text{ s}^{-1}$. Its energy of activation is 209 kJ / mol. Calculate the rate constant of the reaction at 700K.

Key formula

We know that

$$\log k_2 - \log k_1 = \frac{E_a}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

Substitute

$$\log k_2 = \log k_1 + \frac{E_a}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$= \log (1.60 \times 10^{-5}) + \frac{209000 \text{ J mol L}^{-1}}{2.303 \times 8.314 \text{ J mol L}^{-1} \text{ K}^{-1}} \left[\frac{1}{600 \text{ K}} - \frac{1}{700 \text{ K}} \right]$$

$$\log k_2 = -4.796 + 2.599 = -2.197$$

Conclusion

$$k_2 = 6.36 \times 10^{-3} \text{ s}^{-1} \quad \boxed{k_2 = 6.36 \times 10^{-3} \text{ s}^{-1}} //$$

Part 2 · Additional Questions

Q

1

Average rate calculation

(())

For the reaction $R \rightarrow P$, the concentration of a reactant changes from 0.03M to 0.02M in 25 minutes.

Calculate the average rate of reaction using units of time both in minutes and seconds.

Key formula

$$= - \frac{\Delta[R]}{\Delta t}$$



Substitute

$$\begin{aligned} &= - \frac{[R]_2 - [R]_1}{t_2 - t_1} \\ &= - \frac{0.02 - 0.03}{25} \text{ M min}^{-1} \\ &= - \frac{-0.01}{25} \text{ M min}^{-1} \\ &= 4 \times 10^{-4} \text{ M min}^{-1} \\ &= \frac{4 \times 10^{-4}}{60} \text{ M s}^{-1} \\ &= 6.67 \times 10^{-6} \text{ M s}^{-1} \end{aligned}$$

$$4 \times 10^{-4} \text{ M min}^{-1} = 6.67 \times 10^{-6} \text{ M s}^{-1} //$$

Q

2

Rate from concentration change



In a reaction, $2A \rightarrow \text{Products}$, the concentration of A decreases from 0.5 mol L⁻¹ to 0.4 mol L⁻¹ in 10 minutes. Calculate the rate during this interval?

Key formula

$$\text{Average rate} = - \frac{1}{2} \frac{\Delta[A]}{\Delta t}$$



Substitute

$$\begin{aligned} &= - \frac{1}{2} \frac{[A]_2 - [A]_1}{t_2 - t_1} \\ &= - \frac{1}{2} \frac{0.4 - 0.5}{10} \\ &= - \frac{1}{2} \frac{-0.1}{10} \\ &= 0.005 \text{ mol L}^{-1} \text{ min}^{-1} \\ &= 5 \times 10^{-3} \text{ M min}^{-1} \end{aligned}$$

$$5 \times 10^{-3} \text{ M min}^{-1} //$$

Q

3

Order from rate law



For a reaction, $A + B \rightarrow \text{Product}$; the rate law is given by, $r = k[A]^{1/2}[B]^2$. What is the order of the reaction?

Conclusion

$$= \frac{1}{2} + 2$$

The order of the reaction

$$= 2 \frac{1}{2}$$

$$\text{The order of the reaction} = 2.5 //$$

= 2.5

Q

4

Second order
rate effect



The conversion of molecules X to Y follows second order kinetics.
If concentration of X is increased to three times how will it affect the rate of formation of Y ?

Key formula

The reaction $X \rightarrow Y$ follows second order kinetics.

So the rate equation is:

$$\text{Rate} = k[X]^2 \quad (1)$$

Substitute

Let $[X] = a \text{ mol L}^{-1}$. Then equation (1) becomes:

$$\text{Rate}_1 = k \cdot (a)^2$$

$$= k a^2$$

If the concentration of X is increased to three times, $[X] = 3 a \text{ mol L}^{-1}$

Now the rate equation is:

$$\text{Rate} = k(3 a)^2$$

$$= 9(k a^2)$$

Conclusion

So the rate of formation increases by 9 times.

The rate of formation of Y will increase by 9 times. ///

Q

5

First order
half-life time

(())

A first order reaction has a rate constant $1.15 \times 10^{-3} \text{ s}^{-1}$. How long will 5 g of this reactant take to reduce to 3 g?

Given

From the question, we know:

Initial amount = 5 g

Final concentration = 3 g

Rate constant = $1.15 \times 10^{-3} \text{ s}^{-1}$

Substitute

We know that for a 1st order reaction,

$$t = \frac{2.303}{k} \log \frac{[R]_0}{[R]}$$

$$= \frac{2.303}{1.15 \times 10^{-3}} \log \frac{5}{3}$$

$$= \frac{2.303}{1.15 \times 10^{-3}} \times 0.2219$$

$$= 444.38 \text{ s}$$

$$= 444 \text{ s (approx)}$$

$$t = 444 \text{ s (approx)} ///$$

Q

6

Rate constant
from half-life

Time required to decompose $\text{SO}_2 \text{Cl}_2$ to half of its initial amount is 60 minutes. If the decomposition is a first order reaction, calculate the rate constant of the reaction.

Key formula

We know that for a 1st order reaction,

$$t_{1/2} = \frac{0.693}{k}$$

Given

It is given that $t_{1/2} = 60 \text{ min}$ 

Substitute

$$\therefore k = \frac{0.693}{t_{1/2}}$$

$$= \frac{0.693}{60}$$

$$= 0.01155 \text{ min}^{-1}$$

$$= 1.155 \text{ min}^{-1}$$

$$\text{Or } k = 1.925 \times 10^{-4} \text{ s}^{-1}$$

$$k = 1.925 \times 10^{-4} \text{ s}^{-1} //$$

Q

7

Temperature
effect on rate
constant

What will be the effect of temperature on rate constant ?

Conclusion

The rate constant of a reaction nearly doubles with a 10° rise in temperature. The exact dependence of reaction rate on temperature is given by the Arrhenius equation:

$$k = Ae^{-E_a / RT}$$

Where,

A is the Arrhenius factor or the frequency factor

T is the temperature

R is the gas constant

 E_a is the activation energy

Q

8

Activation
energy
calculation

The rate of the chemical reaction doubles for an increase of 10K in absolute temperature from 298K. Calculate E_a .

Given

It is given that $T_1 = 298 \text{ K}$

$$\therefore T_2 = (298 + 10) \text{ K}$$

$$= 308 \text{ K}$$

We also know the rate of the reaction doubles when temperature increases by 10°.

So let us take $k_1 = k$ and $k_2 = 2k$.

$$\text{Also, } R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

Key formula

Now substitute these values in the equation:



$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Substitute

We get:

$$\log \frac{2k}{k} = \frac{E_a}{2.303 \times 8.314} \left[\frac{10}{298 \times 308} \right]$$

$$\Rightarrow \log 2 = \frac{E_a}{2.303 \times 8.314} \left[\frac{10}{298 \times 308} \right]$$

$$E_a = 52.9 \text{ kJ mol}^{-1} //$$

$$\Rightarrow E_a = \frac{2.303 \times 8.314 \times 298 \times 308 \times \log 2}{10}$$

$$= 52897.78 \text{ J mol}^{-1}$$

$$= 52.9 \text{ kJ mol}^{-1}$$

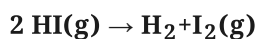
Q

9

Fraction of molecules with activation energy

(())

The activation energy for the reaction



is $209.5 \text{ kJ mol}^{-1}$ at 581 K . Calculate the fraction of molecules of reactants having energy equal to or greater than activation energy?

Given

In the given case:

$$E_a = 209.5 \text{ kJ mol}^{-1} = 209500 \text{ J mol}^{-1}$$

$$T = 581 \text{ K}$$

$$R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$$

Key formula

Now, the fraction of molecules of reactants having energy equal to or greater than activation energy is given as:

$$x = e^{-E_a / R T}$$

$$\Rightarrow \ln x = -E_a / R T$$

$$\Rightarrow \log x = - \frac{E_a}{2.303 R T}$$

Substitute

$$\Rightarrow \log x = \frac{209500 \text{ J mol}^{-1}}{2.303 \times 8.314 \text{ JK}^{-1} \text{ mol}^{-1} \times 581} = 18.8323$$

Now, $x = \text{Antilog}(18.8323)$

$$= \text{Anti log } 19.1677$$

$$= 1.471 \times 10^{-19}$$

$$x = 1.471 \times 10^{-19} //$$

Q

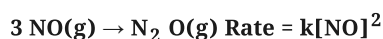
1

Order and
rate constant
dimensions

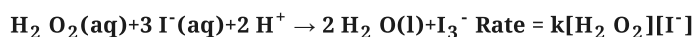


From the rate expression for the following reactions, determine their order of reaction and the dimensions of the rate constants.

Part (i)



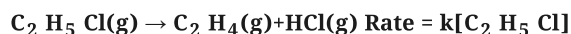
Part (ii)



Part (iii)



Part (iv)



Given

$$(i) \text{ Given rate} = k[\text{NO}]^2$$

Therefore, order of the reaction = 2

Key formula

Dimension of

$$k = \frac{\text{Rate}}{[\text{NO}]^2}$$

Substitute

$$\begin{aligned} &= \frac{\text{mol L}^{-1} \text{ s}^{-1}}{(\text{mol L}^{-1})^2} \\ &= \frac{\text{mol L}^{-1} \text{ s}^{-1}}{\text{mol}^2 \text{ L}^{-2}} \\ &= \text{L mol}^{-1} \text{ s}^{-1} \end{aligned}$$

Given

$$(ii) \text{ Given rate} = k[\text{H}_2 \text{ O}_2][\text{I}^-]$$

Therefore, order of the reaction = 2

Key formula

Dimension of

$$k = \frac{\text{Rate}}{[\text{H}_2 \text{ O}_2][\text{I}^-]}$$

Substitute

$$\begin{aligned} &= \frac{\text{mol L}^{-1} \text{ s}^{-1}}{(\text{mol L}^{-1})(\text{mol L}^{-1})} \\ &= \text{L mol}^{-1} \text{ s}^{-1} \end{aligned}$$

Given

$$(iii) \text{ Given rate} = k[\text{CH}_3 \text{ CHO}]^{3/2}$$

$$\text{Therefore, order of reaction} = \frac{3}{2}$$

Key formula

Dimension of

$$k = \frac{\text{Rate}}{[\text{CH}_3 \text{ CHO}]^{\frac{3}{2}}}$$

Substitute

$$\begin{aligned}
 &= \frac{\text{mol L}^{-1} \text{ s}^{-1}}{(\text{mol L}^{-1})^{\frac{3}{2}}} \\
 &= \frac{\text{mol L}^{-1} \text{ s}^{-1}}{\text{mol}^{\frac{3}{2}} \text{ L}^{-\frac{3}{2}}} \\
 &= \text{L}^{\frac{1}{2}} \text{ mol}^{-\frac{1}{2}} \text{ s}^{-1}
 \end{aligned}$$

Given

(iv) Given rate = $k[\text{C}_2\text{H}_5\text{Cl}]$ Therefore, order of the reaction = 1

Key formula

Dimension of

$$k = \frac{\text{Rate}}{[\text{C}_2\text{H}_5\text{Cl}]}$$

Substitute

$$\begin{aligned}
 &= \frac{\text{mol L}^{-1} \text{ s}^{-1}}{\text{mol L}^{-1}} \\
 &= \text{s}^{-1}
 \end{aligned}$$

(i) order = 2, k in $\text{L mol}^{-1} \text{ s}^{-1}$; (ii) order = 2, k in $\text{L mol}^{-1} \text{ s}^{-1}$; (iii)
 order = $\frac{3}{2}$, k in $\text{L}^{\frac{1}{2}} \text{ mol}^{-\frac{1}{2}} \text{ s}^{-1}$; (iv) order = 1, k in s^{-1}

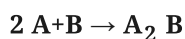
Q

2

Initial rate calculation



For the reaction:



the rate = $k[\text{A}][\text{B}]^2$ with $k = 2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$. Calculate the initial rate of the reaction when $[\text{A}] = 0.1 \text{ mol L}^{-1}$, $[\text{B}] = 0.2 \text{ mol L}^{-1}$. Calculate the rate of reaction after $[\text{A}]$ is reduced to 0.06 mol L^{-1} .

Given

The initial rate of the reaction is

Substitute

$$\begin{aligned}
 \text{Rate} &= k[\text{A}][\text{B}]^2 \\
 &= (2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1})(0.1 \text{ mol L}^{-1})(0.2 \text{ mol L}^{-1})^2 \\
 &= 8.0 \times 10^{-9} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}
 \end{aligned}$$

When $[\text{A}]$ is reduced from 0.1 mol L^{-1} to 0.06 mol L^{-1} , the concentration of A reacted = $(0.1 - 0.06) \text{ mol L}^{-1} = 0.04 \text{ mol L}^{-1}$

Therefore, concentration of B reacted

$$= \frac{1}{2} \times 0.04 \text{ mol L}^{-1}$$

$$= 0.02 \text{ mol L}^{-1}$$

Then, concentration of B available, $[B] = (0.2 - 0.02) \text{ mol L}^{-1}$

$$= 0.18 \text{ mol L}^{-1}$$

After $[A]$ is reduced to 0.06 mol L^{-1} , the rate of the reaction is given by,

$$\text{Rate} = k[A][B]^2$$

$$= (2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1})(0.06 \text{ mol L}^{-1})(0.18 \text{ mol L}^{-1})^2$$

$$= 3.9 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$$

Initial rate $= 8.0 \times 10^{-9} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$; after $[A]$ falls to 0.06 mol L^{-1} ,
rate $= 3.9 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$

///

Q

3

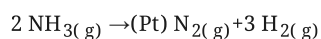
Zero order
rate of
production



The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of production of N_2 and H_2 if $k = 2.5 \times 10^{-4} \text{ mol}^{-1} \text{ L s}^{-1}$?

Given

The decomposition of NH_3 on platinum surface is represented by the following equation.



Key formula



Therefore,

$$\text{Rate} = -\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt}$$

Substitute

However, it is given that the reaction is of zero order.

Therefore,

$$-\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt} = k$$

$$= 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

Conclusion

Therefore, the rate of production of N_2 is

$$\frac{d[\text{N}_2]}{dt} = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

And, the rate of production of H_2 is

$$\frac{d[\text{H}_2]}{dt} = 3 \times 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$= 7.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

**Rate of production of $\text{N}_2 = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$; rate of
 production of $\text{H}_2 = 7.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$**

///

Q

4

Units of rate
and rate
constant

()

The decomposition of dimethyl ether leads to the formation of CH_4 , H_2 and CO and the reaction rate is given by

$$\text{Rate} = k[\text{CH}_3\text{OCH}_3]^{3/2}$$

The rate of reaction is followed by increase in pressure in a closed vessel, so the rate can also be expressed in terms of the partial pressure of dimethyl ether, i.e.,

$$\text{Rate} = k(p_{\text{CH}_3\text{OCH}_3})^{3/2}$$

If the pressure is measured in bar and time in minutes, then what are the units of rate and rate constants?

Given

If pressure is measured in bar and time in minutes, then

Unit of rate = bar min^{-1}

Key formula

$$\text{Rate} = k(p_{\text{CH}_3\text{OCH}_3})^{3/2}$$



$$\Rightarrow k = \frac{\text{Rate}}{(p_{\text{CH}_3\text{OCH}_3})^{3/2}}$$

Substitute

Therefore, unit of rate constants

$$\begin{aligned} (k) &= \frac{\text{bar min}^{-1}}{\text{bar}^{3/2}} \\ &= \text{bar}^{-1/2} \text{ min}^{-1} \end{aligned}$$

Unit of rate = bar min^{-1} ; unit of rate constant (k) = $\text{bar}^{-1/2} \text{ min}^{-1}$

///

Q

5

Factors
affecting
reaction rate

Mention the factors that affect the rate of a chemical reaction.

Conclusion

The factors that affect the rate of a reaction are as follows.

- (i) Concentration of reactants (pressure in case of gases)
- (ii) Temperature
- (iii) Presence of a catalyst

Q

6

Second order
concentration
effect

A reaction is second order with respect to a reactant.

How is the rate of reaction affected if the concentration of the reactant is

Part (i)

doubled

Part (ii)

reduced to half?

Given

Let the concentration of the reactant be $[A] = a$

Key formula

$$\begin{aligned} \text{Rate of reaction, } R &= k[A]^2 \\ &= k a^2 \end{aligned}$$

**Substitute**

(i) If the concentration of the reactant is doubled, i.e. $[A] = 2a$, then the rate of the reaction would be

$$R' = k(2a)^2$$

$$= 4ka^2$$

$$= 4R$$

Conclusion

Therefore, the rate of the reaction would increase by 4 times.

Substitute

(ii) If the concentration of the reactant is reduced to half, i.e. $[A] = \frac{1}{2}a$, then the

rate of the reaction would be

$$R'' = k\left(\frac{1}{2}a\right)^2$$

$$= \frac{1}{4}ka^2$$

$$= \frac{1}{4}R$$

Conclusion

Therefore, the rate of the reaction would be reduced to $\frac{1}{4}$ th.

(i) Rate increases by 4 times; (ii) rate is reduced to $\frac{1}{4}$ th //

Q**7**

**Temperature
dependence
of rate
constant**



What is the effect of temperature on the rate constant of a reaction? How can this effect of temperature on rate constant be represented quantitatively?

Given

For most chemical reactions, the rate constant nearly doubles when the temperature rises by 10° .

The Arrhenius equation expresses this temperature effect quantitatively:

Key formula

$$k = A e^{-E_a / RT}$$

**Conclusion**

where k is the rate constant,

A is the Arrhenius factor (or frequency factor),

R is the gas constant,

T is the temperature, and

E_a is the energy of activation for the reaction

Q

8

Pseudo first
order
average rate

(()

In a pseudo first order reaction in water, the following results were obtained:

t / s	0	30	60	90
[A] / mol L ⁻¹	0.55	0.31	0.17	0.085

Calculate the average rate of reaction between the time interval 30 to 60 seconds.

Substitute

(i) Average rate of reaction between the time interval, 30 to 60 seconds, = -

$$\begin{aligned}
 & \frac{\Delta[\text{Ester}]}{\Delta t} \\
 &= \frac{0.31 - 0.17}{60 - 30} \\
 &= \frac{0.14}{30} \\
 &= 4.67 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}
 \end{aligned}$$

Key formula

(ii) For a pseudo first order reaction,

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

Substitute

For

$$k_1 = \frac{2.303}{30} \log \frac{0.55}{0.31} \quad t = 30 \text{ s,}$$

For t = 60 s,

$$\begin{aligned}
 k_2 &= \frac{2.303}{60} \log \frac{0.55}{0.17} \\
 &= 1.957 \times 10^{-2} \text{ s}^{-1}
 \end{aligned}$$

For t = 90 s,

$$\begin{aligned}
 k_3 &= \frac{2.303}{90} \log \frac{0.55}{0.085} \\
 &= 2.075 \times 10^{-2} \text{ s}^{-1}
 \end{aligned}$$

Conclusion

Then, average rate constant,

$$\begin{aligned}
 k &= \frac{k_1 + k_2 + k_3}{3} \\
 &= \frac{(1.911 \times 10^{-2}) + (1.957 \times 10^{-2}) + (2.075 \times 10^{-2})}{3} \\
 &= 1.98 \times 10^{-2} \text{ s}^{-1}
 \end{aligned}$$

(i) Average rate (30–60 s) = $4.67 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$; (ii) average pseudo first order $k = 1.98 \times 10^{-2} \text{ s}^{-1}$

///

Q

9

Mixed order
rate equation

A reaction is first order in A and second order in B.

Part (i)

Write the differential rate equation.

Part (ii)

How is the rate affected on increasing the concentration of B three times?

Part (iii)

How is the rate affected when the concentrations of both A and B are doubled?

Given

(i) The differential rate equation will be

Key formula

$$-\frac{d[R]}{dt} = k[A][B]^2$$



Substitute

(ii) If the concentration of B is increased three times, then

$$\begin{aligned} -\frac{d[R]}{dt} &= k[A][3B]^2 \\ &= 9 \cdot k[A][B]^2 \end{aligned}$$

Conclusion

Therefore, the rate of reaction will increase 9 times.

Substitute

(iii) When the concentrations of both A and B are doubled,

$$\begin{aligned} -\frac{d[R]}{dt} &= k[2A][2B]^2 \\ &= k[2A][2B]^2 \\ &= 8 \cdot k[A][B]^2 \end{aligned}$$

Conclusion

Therefore, the rate of reaction will increase 8 times.

(i) - $\frac{d[R]}{dt} = k[A][B]^2$; (ii) rate increases 9 times; (iii) rate increases 8 times

//

Q

10

Order
determination
from data

In a reaction between A and B, the initial rate of reaction (r_0) was measured for different initial concentrations of A and B as given below:

A / mol L ⁻¹	0.20	0.20	0.40
B / mol L ⁻¹	0.30	0.10	0.05
r_0 / mol L ⁻¹ s ⁻¹	5.07×10^{-5}	5.07×10^{-5}	1.43×10^{-4}

What is the order of the reaction with respect to A and B?

Given

Let the order of the reaction with respect to A be x and with respect to B be y.

Therefore,

$$r_0 = k[A]^x[B]^y$$

$$5.07 \times 10^{-5} = k[0.20]^x[0.30]^y$$

$$5.07 \times 10^{-5} = k[0.20]^x[0.10]^y$$

$$1.43 \times 10^{-4} = k[0.40]^x[0.05]^y$$

Substitute

Dividing equation (i) by (ii), we get

$$\frac{5.07 \times 10^{-5}}{5.07 \times 10^{-5}} = \frac{k[0.20]^x[0.30]^y}{k[0.20]^x[0.10]^y}$$

$$\Rightarrow 1 = \frac{[0.30]^y}{[0.10]^y}$$

$$\Rightarrow \left(\frac{0.30}{0.10} \right)^0 = \left(\frac{0.30}{0.10} \right)^y$$

$$\Rightarrow y = 0$$

Dividing equation (iii) by (ii), we get

$$\frac{1.43 \times 10^{-4}}{5.07 \times 10^{-5}} = \frac{k[0.40]^x[0.05]^y}{k[0.20]^x[0.30]^y}$$

$$\Rightarrow \frac{1.43 \times 10^{-4}}{5.07 \times 10^{-5}} = \frac{[0.40]^x}{[0.20]^x} [$$

Since $y = 0$,

$$[0.05]^{y < / sup> = [0.30]^{y < / sup> = 1$$

]

$$\Rightarrow 2.821 = 2^x$$

$$\Rightarrow \log 2.821 = x \log 2 \text{ (Taking log on both sides)}$$

$$\Rightarrow x = \frac{\log 2.821}{\log 2}$$

$$= 1.496$$

$$= 1.5 \text{ (approximately)}$$

Conclusion

Hence, the order of the reaction with respect to A is 1.5 and with respect to B is zero.

Order with respect to A = 1.5; order with respect to B = 0 //

Q

11

Rate law
from kinetic
data



The following results have been obtained during the kinetic studies of the reaction:



Experiment	[A] / mol L ⁻¹	[B] / mol L ⁻¹	Initial rate of formation of D / mol L ⁻¹ min ⁻¹
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

Determine the rate law and the rate constant for the reaction.

Given

Let the order of the reaction with respect to A be x and with respect to B be y .

Therefore, rate of the reaction is given by,

Key formula

$$\text{Rate} = k[\text{A}]^x[\text{B}]^y$$

Substitute

According to the question,

$$6.0 \times 10^{-3} = k[0.1]^x[0.1]^y$$

$$7.2 \times 10^{-2} = k[0.3]^x[0.2]^y$$

$$2.88 \times 10^{-1} = k[0.3]^x[0.4]^y$$

$$2.40 \times 10^{-2} = k[0.4]^x[0.1]^y$$

Dividing equation (iv) by (i), we get

$$\frac{2.40 \times 10^{-2}}{6.0 \times 10^{-3}} = \frac{k[0.4]^x[0.1]^y}{k[0.1]^x[0.1]^y}$$

$$\Rightarrow 4 = \frac{[0.4]^x}{[0.1]^x}$$

$$\Rightarrow 4 = \left(\frac{0.4}{0.1} \right)^x$$

$$\Rightarrow (4)^1 = 4^x$$

$$\Rightarrow x = 1$$

Dividing equation (iii) by (ii), we get

$$\frac{2.88 \times 10^{-1}}{7.2 \times 10^{-2}} = \frac{k[0.3]^x[0.4]^y}{k[0.3]^x[0.2]^y}$$

$$\Rightarrow 4 = \left(\frac{0.4}{0.2} \right)^y$$

$$\Rightarrow 4 = 2^y$$

$$\Rightarrow 2^2 = 2^y$$

$$\Rightarrow y = 2$$

Key formula

Therefore, the rate law is

$$\text{Rate} = k[\text{A}][\text{B}]^2$$

$$k = \frac{\text{Rate}}{[\text{A}][\text{B}]^2}$$

Substitute

From experiment I, we get

$$\begin{aligned} k &= \frac{6.0 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}}{(0.1 \text{ mol L}^{-1})(0.1 \text{ mol L}^{-1})^2} \\ &= 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1} \end{aligned}$$

From experiment II, we get

$$\begin{aligned} k &= \frac{7.2 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}}{(0.3 \text{ mol L}^{-1})(0.2 \text{ mol L}^{-1})^2} \\ &= 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1} \end{aligned}$$

From experiment III, we get

$$k = \frac{2.88 \times 10^{-1} \text{ mol L}^{-1} \text{ min}^{-1}}{(0.3 \text{ mol L}^{-1})(0.4 \text{ mol L}^{-1})^2}$$
$$= 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$$

From experiment IV, we get

$$k = \frac{2.40 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}}{(0.4 \text{ mol L}^{-1})(0.1 \text{ mol L}^{-1})^2}$$
$$= 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$$

Conclusion

Therefore, rate constant, $k = 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$

$$\text{Rate law} = k[A][B]^2; k = 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1} //$$

Q

12

Fill in rate table

(())

The reaction between A and B is first order with respect to A and zero order with respect to B.

Fill in the blanks in the following table:

Experiment	[A] / mol L ⁻¹	[B] / mol L ⁻¹	Initial rate / mol L ⁻¹ min ⁻¹
I	0.1	0.1	2.0×10^{-2}
II	-	0.2	4.0×10^{-2}
III	0.4	0.4	-
IV	-	0.2	2.0×10^{-2}

Given

The given reaction is of the first order with respect to A and of zero order with respect to B.

Therefore, the rate of the reaction is given by,

Key formula

$$\text{Rate} = k[A]^1[B]^0$$

$$\Rightarrow \text{Rate} = k[A]$$

Substitute

From experiment I, we get

$$2.0 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1} = k(0.1 \text{ mol L}^{-1})$$

$$\Rightarrow k = 0.2 \text{ min}^{-1}$$

From experiment II, we get

$$4.0 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1} = 0.2 \text{ min}^{-1}[A]$$

$$\Rightarrow [A] = 0.2 \text{ mol L}^{-1}$$

From experiment III, we get Rate

$$= 0.2 \text{ min}^{-1} \times 0.4 \text{ mol L}^{-1}$$

$$= 0.08 \text{ mol L}^{-1} \text{ min}^{-1}$$

From experiment IV, we get

$$2.0 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1} = 0.2 \text{ min}^{-1}[A]$$

$$\Rightarrow [A] = 0.1 \text{ mol L}^{-1}$$

$k = 0.2 \text{ min}^{-1}$; experiment II $[A] = 0.2 \text{ mol L}^{-1}$; experiment III rate
 $= 0.08 \text{ mol L}^{-1} \text{ min}^{-1}$; experiment IV $[A] = 0.1 \text{ mol L}^{-1}$

Q

13

Half-life
from rate
constant



Calculate the half-life of a first order reaction from their rate constants given below:

Part (i)

200 s^{-1}

Part (ii)

2 min^{-1}

Part (iii)

4 years^{-1}

Substitute

$$\begin{aligned} \text{(i) Half life, } t_{1/2} &= \frac{0.693}{k} \\ &= \frac{0.693}{200 \text{ s}^{-1}} \\ &= 3.47 \text{ s (approximately)} \end{aligned}$$

Key formula

$$\text{(ii) Half life, } t_{1/2} = \frac{0.693}{k}$$



Substitute

$$\begin{aligned} &= \frac{0.693}{2 \text{ min}^{-1}} \\ &= 0.35 \text{ min (approximately)} \end{aligned}$$

Key formula

$$\text{(iii) Half life, } t_{1/2} = \frac{0.693}{k}$$



Substitute

$$\begin{aligned} &= \frac{0.693}{4 \text{ years}^{-1}} \\ &= 0.173 \text{ years (approximately)} \end{aligned}$$

(i) $t_{1/2} = 3.47 \text{ s}$; (ii) $t_{1/2} = 0.35 \text{ min}$; (iii) $t_{1/2} = 0.173 \text{ years}$

Q

14

Radiocarbon
dating



The half-life for radioactive decay of ^{14}C is 5730 years. An archaeological artifact containing wood had only 80% of the ^{14}C found in a living tree. Estimate the age of the sample.

Key formula

Here,

$$\begin{aligned} k &= \frac{0.693}{t_{1/2}} \\ &= \frac{0.693}{5730} \text{ years}^{-1} \end{aligned}$$

Substitute

It is known that,

$$t = \frac{2.303}{k} \log \frac{[R]_0}{[R]}$$

$$= \frac{2.303}{0.693} \log \frac{100}{80}$$

$$= 5730$$

$$= 1845 \text{ years (approximately)}$$

Conclusion

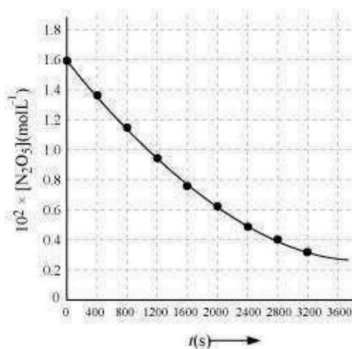
Hence, the age of the sample is 1845 years.

Age of the sample ≈ 1845 years //

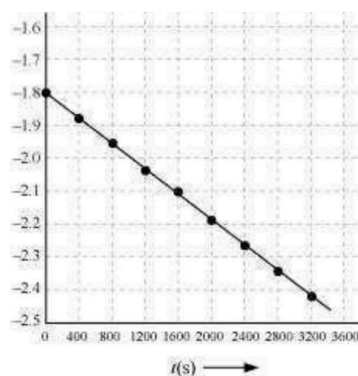
Q

15

N₂O₅ decomposition kinetics

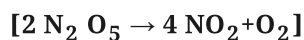


Plot of [N₂O₅] against t



Plot of log [N₂O₅] against t

The experimental data for decomposition of N₂O₅



in gas phase at 318K are given below:

t / s	0	400	800	1200	1600	2000	2400	2800	3200
10 ² × [N ₂ O ₅] / mol L ⁻¹	1.63	1.36	1.14	0.93	0.78	0.64	0.53	0.43	0.35

Part (i)

Plot [N₂O₅] against t.

Part (ii)

Find the half-life period for the reaction.

Part (iii)

Draw a graph between log[N₂O₅] and t.

Part (iv)

What is the rate law?

Part (v)

Calculate the rate constant.

Part (vi)

Calculate the half-life period from k and compare it with (ii).

Given

(ii) Time corresponding to the concentration, $\frac{1.630 \times 10^2}{2} \text{ mol L}^{-1} = 81.5 \text{ mol L}^{-1}$, is the half life.

Conclusion

From the graph, the half life is obtained as 1450 s.

Substitute

(iii)

t(s)	$10^2 \times [\text{N}_2 \text{O}_5] / \text{mol L}^{-1}$	$\log[\text{N}_2 \text{O}_5]$
0	1.63	- 1.79
400	1.36	- 1.87
800	1.14	- 1.94
1200	0.93	- 2.03
1600	0.78	- 2.11
2000	0.64	- 2.19
2400	0.53	- 2.28
2800	0.43	- 2.37
3200	0.35	- 2.46

Given

(iv) The given reaction is of the first order as the plot of $\log [\text{N}_2 \text{O}_5]$ vs t is a straight line.

Key formula

Therefore, the rate law of the reaction is

$$\text{Rate} = k[\text{N}_2 \text{O}_5]$$

Substitute

(v) From the plot, $\log [\text{N}_2 \text{O}_5]$

$$\text{Slope} = \frac{-2.46 - (-1.79)}{3200 - 0}$$

$$= \frac{-0.67}{3200}$$

vs t, we get

Again, slope of the line of the plot of $\log [\text{N}_2 \text{O}_5]$ vs t is given by

$$= -\frac{k}{2.303}$$

Therefore, we get,

$$-\frac{k}{2.303} = -\frac{0.67}{3200}$$

$$\Rightarrow k = 4.82 \times 10^{-4} \text{ s}^{-1}$$

Key formula

(vi) Half-life is given by,

Substitute

$$\begin{aligned}
 t_{1/2} &= \frac{0.693}{k} \\
 &= \frac{0.693}{4.82 \times 10^{-4}} \text{ s} \\
 &= 1.438 \times 10^3 \text{ s} \\
 &= 1438 \text{ s}
 \end{aligned}$$

Conclusion

This value, 1438 s, is very close to the value that was obtained from the graph.

(ii) $t_{1/2}$ (from graph) = 1450 s; (iv) Rate = $k[\text{N}_2\text{O}_5]$; (v)
 $k = 4.82 \times 10^{-4} \text{ s}^{-1}$; (vi) $t_{1/2}$ (from k) = 1438 s

///

Q

16

First order
concentration
reduction
time

(())

The rate constant for a first order reaction is 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to its $1/16^{\text{th}}$ value?

Given

It is known that,

Substitute

$$\begin{aligned}
 t &= \frac{2.303}{k} \log \frac{[\text{R}]_0}{[\text{R}]} \\
 &= \frac{2.303}{60 \text{ s}^{-1}} \log \frac{1}{1/16} \\
 &= \frac{2.303}{60 \text{ s}^{-1}} \log 16 \\
 &= 4.6 \times 10^{-2} \text{ s (approximately)}
 \end{aligned}$$

Conclusion

Hence, the required time is $4.6 \times 10^{-2} \text{ s}$. $t = 4.6 \times 10^{-2} \text{ s}$ ///

Q

17

Radioactive
decay
remaining
amount

⊛

During nuclear explosion, one of the products is ^{90}Sr with half-life of 28.1 years. If $1 \mu\text{g}$ of ^{90}Sr was absorbed in the bones of a newly born baby instead of calcium, how much of it will remain after 10 years and 60 years if it is not lost metabolically.

Key formula

Here,

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{28.1} \text{ y}^{-1}$$

Substitute

It is known that,

$$t = \frac{2.303}{k} \log \frac{[\text{R}]_0}{[\text{R}]}$$

$$\Rightarrow 10 = \frac{2.303}{\frac{0.693}{28.1}} \log \frac{1}{[R]}$$

$$\Rightarrow 10 = \frac{2.303}{\frac{0.693}{28.1}} (-\log [R])$$

$$\Rightarrow \log [R] = - \frac{10 \times 0.693}{2.303 \times 28.1}$$

$$\Rightarrow [R] = \text{antilog}(-0.1071)$$

$$= \text{antilog}(\bar{1}.8929)$$

$$= 0.7814 \mu\text{g}$$

Conclusion

Therefore, 0.7814 μg of ^{90}Sr will remain after 10 years.

Substitute

Again,

$$t = \frac{2.303}{k} \log \frac{[R]_0}{[R]}$$

$$\Rightarrow 60 = \frac{2.303}{\frac{0.693}{28.1}} \log \frac{1}{[R]}$$

$$\Rightarrow \log [R] = - \frac{60 \times 0.693}{2.303 \times 28.1}$$

$$\Rightarrow [R] = \text{antilog}(-0.6425)$$

$$= \text{antilog}(\bar{1}.3575)$$

$$= 0.2278 \mu\text{g}$$

Conclusion

Therefore, 0.2278 μg of ^{90}Sr will remain after 60 years.

After 10 years, 0.7814 μg of ^{90}Sr remains; after 60 years, 0.2278 μg remains

Q

18

First order
completion
time proof



For a first order reaction, show that time required for 99% completion is twice the time required for the completion of 90% of reaction.

Key formula

For a first order reaction, the time required for 99% completion is



Substitute

$$t_1 = \frac{2.303}{k} \log \frac{100}{100-99}$$

$$= \frac{2.303}{k} \log 100$$

$$= 2 \times \frac{2.303}{k}$$

Key formula

For a first order reaction, the time required for 90% completion is

**Substitute**

$$\begin{aligned}
 t_2 &= \frac{2.303}{k} \log \frac{100}{100-90} \\
 &= \frac{2.303}{k} \log 10 \\
 &= \frac{2.303}{k}
 \end{aligned}$$

ConclusionTherefore, $t_1 = 2 t_2$

Hence, the time required for 99% completion of a first order reaction is twice the time required for the completion of 90% of the reaction.

Q**19**

Half-life
from
decomposition
data



A first order reaction takes 40 min for 30% decomposition. Calculate $t_{1/2}$.

Key formula

For a first order reaction,

**Substitute**

$$\begin{aligned}
 t &= \frac{2.303}{k} \log \frac{[R]_0}{[R]} \\
 k &= \frac{2.303}{40 \text{ min}} \log \frac{100}{100-30} \\
 &= \frac{2.303}{40 \text{ min}} \log \frac{10}{7} \\
 &= 8.918 \times 10^{-3} \text{ min}^{-1}
 \end{aligned}$$

$$t_{1/2} \approx 77.7 \text{ min} //$$

Therefore, $t_{1/2}$ of the decomposition reaction is

$$\begin{aligned}
 t_{1/2} &= \frac{0.693}{k} \\
 &= \frac{0.693}{8.918 \times 10^{-3}} \text{ min} \\
 &= 77.7 \text{ min (approximately)}
 \end{aligned}$$

Q**20**

Rate constant
from
pressure
data

(())

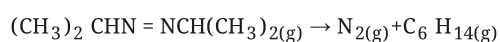
For the decomposition of azoisopropane to hexane and nitrogen at 543 K, the following data are obtained.

t(sec)	P(mm of Hg)
0	35.0
360	54.0
720	63.0

Calculate the rate constant.

Given

The decomposition of azoisopropane to hexane and nitrogen at 543 K is represented by the following equation.



At $t = 0$ P_0 0 0

At $t = t$ $P_0 - p$ p p

Substitute

$$P_t = (P_0 - p) + p + p$$

After time, t , total pressure,

$$\Rightarrow P_t = P_0 + p$$

$$\Rightarrow p = P_t - P_0$$

Therefore, $P_0 - p = P_0 - (P_t - P_0)$

$$= 2P_0 - P_t$$

Key formula

For a first order reaction,



$$k = \frac{2.303}{t} \log \frac{P_0}{P_0 - p}$$
$$= \frac{2.303}{t} \log \frac{P_0}{2P_0 - P_t}$$

Substitute

When $t = 360$ s,

$$k = \frac{2.303}{360 \text{ s}} \log \frac{35.0}{2 \times 35.0 - 54.0}$$
$$= 2.175 \times 10^{-3} \text{ s}^{-1}$$

When $t = 720$ s,

$$k = \frac{2.303}{720 \text{ s}} \log \frac{35.0}{2 \times 35.0 - 63.0}$$
$$= 2.235 \times 10^{-3} \text{ s}^{-1}$$

Conclusion

Hence, the average value of rate constant is

$$k = \frac{(2.175 \times 10^{-3}) + (2.235 \times 10^{-3})}{2} \text{ s}^{-1}$$
$$= 2.21 \times 10^{-3} \text{ s}^{-1}$$

$$\text{Average } k = 2.21 \times 10^{-3} \text{ s}^{-1}$$

///

Textbook variance

Note: There is a slight variation in this answer and the one given in the NCERT textbook.

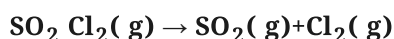
Q

21

Rate from
total
pressure
data



The following data were obtained during the first order thermal decomposition of $\text{SO}_2 \text{Cl}_2$ at a constant volume.

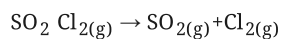


Experiment	Time / s^{-1}	Total pressure / atm
1	0	0.5
2	100	0.6

Calculate the rate of the reaction when total pressure is 0.65 atm.

Given

The thermal decomposition of $\text{SO}_2 \text{Cl}_2$ at a constant volume is represented by the following equation.



At $t = 0$ P_0 0 0

At $t = t$ $P_0 - p$ p p

Substitute

$$P_t = (P_0 - p) + p + p$$

After time, t , total pressure,

$$\Rightarrow P_t = P_0 + p$$

$$\Rightarrow p = P_t - P_0$$

$$P_0 - p = P_0 - (P_t - P_0)$$

Therefore,

$$= 2 P_0 - P_t$$

Key formula

For a first order reaction,



$$k = \frac{2.303}{t} \log \frac{P_0}{P_0 - p}$$

$$= \frac{2.303}{t} \log \frac{P_0}{2 P_0 - P_t}$$

Substitute

When

$$k = \frac{2.303}{100 \text{ s}} \log \frac{0.5}{2 \times 0.5 - 0.6} \quad t = 100 \text{ s},$$

$$= 2.231 \times 10^{-3} \text{ s}^{-1}$$

When $P_t = 0.65 \text{ atm}$,

$$P_0 + p = 0.65$$

$$\Rightarrow p = 0.65 - P_0$$

$$= 0.65 - 0.5$$

$$= 0.15 \text{ atm}$$

Therefore, when the total pressure is 0.65 atm, pressure of SO_2Cl_2 is

$$p_{\text{SO}_2\text{Cl}_2} = P_0 - p$$

$$= 0.5 - 0.15$$

$$= 0.35 \text{ atm}$$

Conclusion

Therefore, the rate of reaction, when total pressure is 0.65 atm, is given by,

$$\text{Rate} = k(p_{\text{SO}_2\text{Cl}_2})$$

$$= (2.23 \times 10^{-3} \text{ s}^{-1})(0.35 \text{ atm})$$

$$= 7.8 \times 10^{-4} \text{ atm s}^{-1}$$

$$\text{Rate (at total pressure 0.65 atm)} = 7.8 \times 10^{-4} \text{ atm s}^{-1}$$

///

Q

22

Arrhenius
plot and
prediction

The rate constant for the decomposition of N_2O_5 at various temperatures is given below:

T / °C	0	20	40	60	80
$10^5 \times k / \text{s}^{-1}$	0.0787	1.70	25.7	178	2140

Draw a graph between $\ln k$ and $1/T$ and calculate the values of A and E_a . Predict the rate constant at 30° and 50°C.

Given

From the given data, we get

T / °C	0	20	40	60	80
T / K	273	293	313	333	353
$10^3 \times \frac{1}{T} (\text{K}^{-1}) \rightarrow$	3.66×10^{-3}	3.41×10^{-3}	3.19×10^{-3}	3.0×10^{-3}	2.83×10^{-3}
$10^5 \times k / \text{s}^{-1}$	0.0787	1.70	25.7	178	2140
$\ln k$	-7.147	-4.075	-1.359	-0.577	3.063

Substitute

Slope of the line,

$$\frac{y_2 - y_1}{x_2 - x_1} = -12.301 \text{ K}$$

According to Arrhenius equation,

$$\text{Slope} = - \frac{E_a}{R}$$

$$\Rightarrow E_a = - \text{Slope} \times R$$

$$= -(-12.301 \text{ K}) \times (8.314 \text{ JK}^{-1} \text{ mol}^{-1})$$

$$= 102.27 \text{ kJ mol}^{-1}$$

Key formula

Again,



$$\ln k = \ln A - \frac{E_a}{R T}$$

$$\ln A = \ln k + \frac{E_a}{R T}$$

Substitute

When T = 273 K,

$$\ln k = -7.147$$

Then,

$$\ln A = -7.147 + \frac{102.27 \times 10^3}{8.314 \times 273}$$

$$= 37.911$$

Conclusion

Therefore, $A = 2.91 \times 10^6$

Substitute

When T = 30+273 K = 303 K,

$$\frac{1}{T} = 0.0033 \text{ K} = 3.3 \times 10^{-3} \text{ K}$$

$$\text{Then, at } \frac{1}{T} = 3.3 \times 10^{-3} \text{ K,}$$

$$\ln k = -2.8$$

Conclusion

$$\text{Therefore, } k = 6.08 \times 10^{-2} \text{ s}^{-1}$$

Substitute

$$\text{Again, when } T = 50 + 273 \text{ K} = 323 \text{ K}$$

$$\frac{1}{T} = 0.0031 \text{ K} = 3.1 \times 10^{-3} \text{ K}$$

$$\text{Then, at } \frac{1}{T} = 3.1 \times 10^{-3} \text{ K}$$

$$\ln k = -0.5$$

Conclusion

$$\text{Therefore, } k = 0.607 \text{ s}^{-1}$$

$$E_a = 102.27 \text{ kJ mol}^{-1}; A = 2.91 \times 10^6; k(30^\circ\text{C}) = 6.08 \times 10^{-2} \text{ s}^{-1};$$

$$k(50^\circ\text{C}) = 0.607 \text{ s}^{-1}$$

///

Q

23

Pre-exponential factor calculation



The rate constant for the decomposition of hydrocarbons is $2.418 \times 10^{-5} \text{ s}^{-1}$ at 546 K. If the energy of activation is 179.9 kJ / mol, what will be the value of pre-exponential factor.

Given

$$k = 2.418 \times 10^{-5} \text{ s}^{-1}$$

$$T = 546 \text{ K}$$

$$E_a = 179.9 \text{ kJ mol}^{-1} = 179.9 \times 10^3 \text{ J mol}^{-1}$$

Substitute

According to the Arrhenius equation,

$$k = Ae^{-E_a / RT}$$

$$\Rightarrow \ln k = \ln A - \frac{E_a}{RT}$$

$$\Rightarrow \log k = \log A - \frac{E_a}{2.303 RT}$$

$$\Rightarrow \log A = \log k + \frac{E_a}{2.303 RT}$$

$$= \log (2.418 \times 10^{-5} \text{ s}^{-1}) + \frac{179.9 \times 10^3 \text{ J mol}^{-1}}{2.303 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 546 \text{ K}}$$

$$= (0.3835 - 5) + 17.2082$$

$$= 12.5917$$

Conclusion

Therefore, A = antilog (12.5917)

$$= 3.9 \times 10^{12} \text{ s}^{-1} \text{ (approximately)}$$

$$A = 3.9 \times 10^{12} \text{ s}^{-1} //$$

Q

24

Concentration remaining after time

()

Consider a certain reaction $A \rightarrow \text{Products}$ with $k = 2.0 \times 10^{-2} \text{ s}^{-1}$. Calculate the concentration of A remaining after 100 s if the initial concentration of A is 1.0 mol L^{-1} .

Given

$$k = 2.0 \times 10^{-2} \text{ s}^{-1} \quad t = 100 \text{ s}$$

$$[A]_0 = 1.0 \text{ mol L}^{-1}$$

Since the unit of k is s^{-1} , the given reaction is a first order reaction.

Key formula

Therefore,



$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

Substitute

$$\Rightarrow 2.0 \times 10^{-2} \text{ s}^{-1} = \frac{2.303}{100 \text{ s}} \log \frac{1.0}{[A]}$$

$$\Rightarrow 2.0 \times 10^{-2} \text{ s}^{-1} = \frac{2.303}{100 \text{ s}} (-\log [A])$$

$$\Rightarrow -\log [A] = \frac{2.0 \times 10^{-2} \times 100}{2.303}$$

$$\Rightarrow [A] = \text{anti log} \left(-\frac{2.0 \times 10^{-2} \times 100}{2.303} \right)$$

$$= 0.135 \text{ mol L}^{-1} \text{ (approximately)}$$

Conclusion

Hence, the remaining concentration of A is 0.135 mol L^{-1} .

$$[A] = 0.135 \text{ mol L}^{-1} //$$

Q

25

Fraction remaining after decay

()

Sucrose decomposes in acid solution into glucose and fructose according to the first order rate law, with $t_{1/2} = 3.00$ hours. What fraction of sample of sucrose remains after 8 hours ?

Key formula

For a first order reaction,



$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

Given

It is given that, $t_{1/2} = 3.00$ hours

Substitute

Therefore,

$$k = \frac{0.693}{t_{1/2}}$$

$$= \frac{0.693}{3} \text{ h}^{-1}$$

$$= 0.231 \text{ h}^{-1}$$

$$\begin{aligned}
 \text{Then, } 0.231 \text{ h}^{-1} &= \frac{2.303}{8 \text{ h}} \log \frac{[R]_0}{[R]} \\
 \Rightarrow \log \frac{[R]_0}{[R]} &= \frac{0.231 \text{ h}^{-1} \times 8 \text{ h}}{2.303} \\
 \Rightarrow \frac{[R]_0}{[R]} &= \text{antilog}(0.8024) \\
 \Rightarrow \frac{[R]_0}{[R]} &= 6.3445 \\
 \Rightarrow \frac{[R]}{[R]_0} &= 0.1576 \text{ (approx)} \\
 &= 0.158
 \end{aligned}$$

Conclusion

Hence, the fraction of sample of sucrose that remains after 8 hours is 0.158 .

Fraction of sucrose remaining after 8 hours = 0.158 ///

Q

26

Activation energy from Arrhenius form



The decomposition of hydrocarbon follows the equation

$$k = (4.5 \times 10^{11} \text{ s}^{-1}) e^{-28000 \text{ K} / T}$$

Calculate E_a .

Given

The given equation is $k = (4.5 \times 10^{11} \text{ s}^{-1}) e^{-28000 \text{ K} / T}$ (i)

Key formula

The Arrhenius equation is

$$k = Ae^{-E_a / RT} \text{ (ii)}$$

Substitute

Comparing equations (i) and (ii), we get

$$\frac{E_a}{RT} = \frac{28000 \text{ K}}{T}$$

$$\Rightarrow E_a = R \times 28000 \text{ K}$$

$$= 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 28000 \text{ K}$$

$$= 232792 \text{ J mol}^{-1}$$

$$= 232.792 \text{ kJ mol}^{-1}$$

$$E_a = 232.792 \text{ kJ mol}^{-1} ///$$

Q

27

E_a and temperature from rate equation



The rate constant for the first order decomposition of H_2O_2 is given by the following equation:

$$\log k = 14.34 - 1.25 \times 10^4 \text{ K} / T$$

Calculate E_a for this reaction and at what temperature will its half-period be 256 minutes?

Key formula

The Arrhenius equation is

$$k = Ae^{-E_a / RT}$$

$$\Rightarrow \ln k = \ln A - \frac{E_a}{R T}$$

$$\Rightarrow \ln k = \log A - \frac{E_a}{R T}$$

$$\Rightarrow \log k = \log A - \frac{E_a}{2.303 R T}$$

Given

The given equation is

$$\log k = 14.34 - 1.25 \times 10^4 \text{ K} / T$$

Substitute

Comparing equations (i) and (ii), we get

$$\frac{E_a}{2.303 R T} = \frac{1.25 \times 10^4 \text{ K}}{T}$$

$$\Rightarrow E_a = 1.25 \times 10^4 \text{ K} \times 2.303 \times R$$

$$= 1.25 \times 10^4 \text{ K} \times 2.303 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$= 239339.3 \text{ J mol}^{-1} \text{ (approximately)}$$

$$= 239.34 \text{ kJ mol}^{-1}$$

Also, when $t_{1/2} = 256$ minutes,

$$k = \frac{0.693}{t_{1/2}}$$

$$= \frac{0.693}{256}$$

$$= 2.707 \times 10^{-3} \text{ min}^{-1}$$

$$= 4.51 \times 10^{-5} \text{ s}^{-1}$$

It is also given that, $\log k = 14.34 - 1.25 \times 10^4 \text{ K} / T$

$$\Rightarrow \log (4.51 \times 10^{-5}) = 14.34 - \frac{1.25 \times 10^4 \text{ K}}{T}$$

$$\Rightarrow \log (0.654 \times 10^{-5}) = 14.34 - \frac{1.25 \times 10^4 \text{ K}}{T}$$

$$\Rightarrow \frac{1.25 \times 10^4 \text{ K}}{T} = 18.686$$

$$\Rightarrow T = \frac{1.25 \times 10^4 \text{ K}}{18.686}$$

$$= 668.95 \text{ K}$$

$$= 669 \text{ K (approximately)}$$

$$E_a = 239.34 \text{ kJ mol}^{-1}; T = 669 \text{ K (for } t_{1/2} = 256 \text{ min)}$$

///

Q

28

Temperature
from
Arrhenius
equation

(())

The decomposition of A into product has value of k as $4.5 \times 10^3 \text{ s}^{-1}$ at 10°C and energy of activation 60 kJ mol^{-1} . At what temperature would k be $1.5 \times 10^4 \text{ s}^{-1}$?

Key formula

Using the Arrhenius equation, we get



$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

Given

$$\text{Also, } k_1 = 4.5 \times 10^3 \text{ s}^{-1}$$

$$T_1 = 273 + 10 = 283 \text{ K}$$

$$k_2 = 1.5 \times 10^4 \text{ s}^{-1}$$

$$E_a = 60 \text{ kJ mol}^{-1} = 6.0 \times 10^4 \text{ J mol}^{-1}$$

Substitute

Then,

$$\log \frac{1.5 \times 10^4}{4.5 \times 10^3} = \frac{6.0 \times 10^4 \text{ J mol}^{-1}}{2.303 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}} \left(\frac{T_2 - 283}{283 T_2} \right)$$

$$\Rightarrow 0.5229 = 3133.627 \left(\frac{T_2 - 283}{283 T_2} \right)$$

$$\Rightarrow \frac{0.5229 \times 283 T_2}{3133.627} = T_2 - 283$$

$$\Rightarrow 0.0472 T_2 = T_2 - 283$$

$$\Rightarrow 0.9528 T_2 = 283$$

$$\Rightarrow T_2 = 297.019 \text{ K (approximately)}$$

$$= 297 \text{ K}$$

$$= 24^\circ \text{C}$$

Conclusion

Hence, k would be $1.5 \times 10^4 \text{ s}^{-1}$ at 24°C . $T_2 = 297 \text{ K} = 24^\circ \text{C}$ //

Textbook variance

Note: There is a slight variation in this answer and the one given in the NCERT textbook.

Q

29

k and E_a
from
completion
times

(())

The time required for 10% completion of a first order reaction at 298 K is equal to that required for its 25% completion at 308 K . If the value of A is $4 \times 10^{10} \text{ s}^{-1}$. Calculate k at 318 K and E_a .

Key formula

For a first order reaction,



$$t = \frac{2.303}{k} \log \frac{a}{a-x}$$

Substitute

At 298 K ,

$$t = \frac{2.303}{k} \log \frac{100}{90}$$

$$= \frac{0.1054}{k}$$

At 308 K ,

$$t' = \frac{2.303}{k'} \log \frac{100}{75}$$

$$= \frac{2.2877}{k'}$$

According to the question,

$$t = t'$$

$$\Rightarrow \frac{0.1054}{k} = \frac{0.2877}{k'}$$

$$\Rightarrow \frac{k'}{k} = 2.7296$$

From the Arrhenius equation, we get

$$\log \frac{k'}{k} = \frac{E_a}{2.303 R} \left(\frac{T' - T}{T T'} \right)$$

$$\log (2.7296) = \frac{E_a}{2.303 \times 8.314} \left(\frac{308 - 298}{298 \times 308} \right)$$

$$E_a = \frac{2.303 \times 8.314 \times 298 \times 308 \times \log (2.7296)}{308 - 298}$$

$$= 76640.096 \text{ J mol}^{-1}$$

$$= 76.64 \text{ kJ mol}^{-1}$$

Given

To calculate k at 318 K ,

It is given that, $A = 4 \times 10^{10} \text{ s}^{-1}$, $T = 318 \text{ K}$

Substitute

Again, from the Arrhenius equation, we get

$$\log k = \log A - \frac{E_a}{2.303 R T}$$

$$= \log (4 \times 10^{10}) - \frac{76.64 \times 10^3}{2.303 \times 8.314 \times 318}$$

$$= (0.6021 + 10) - 12.5876$$

$$= -1.9855$$

Conclusion

Therefore,

$$k = \text{Antilog}(-1.9855)$$

$$= 1.034 \times 10^{-2} \text{ s}^{-1}$$

$$E_a = 76.64 \text{ kJ mol}^{-1}; k \text{ (at 318 K)} = 1.034 \times 10^{-2} \text{ s}^{-1} //$$

Q**30**

Activation
energy from
rate
quadrupling



The rate of a reaction quadruples when the temperature changes from 293 K to 313 K.

Calculate the energy of activation of the reaction assuming that it does not change with temperature.

Key formula

Using the Arrhenius equation, we get



$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

Given

It is given that, $k_2 = 4 k_1$

$T_1 = 293 \text{ K}$

$T_2 = 313 \text{ K}$

Substitute

$$\text{Therefore, } \log \frac{4 k_1}{k_2} = \frac{E_a}{2.303 \times 8.314} \left(\frac{313 - 293}{293 \times 313} \right)$$

$$\Rightarrow 0.6021 = \frac{20 \times E_a}{2.303 \times 8.314 \times 293 \times 313}$$

$$\Rightarrow E_a = \frac{0.6021 \times 2.303 \times 8.314 \times 293 \times 313}{20}$$

$$= 52863.33 \text{ J mol}^{-1}$$

$$= 52.86 \text{ kJ mol}^{-1}$$

Conclusion

Hence, the required energy of activation is $52.86 \text{ kJ mol}^{-1}$.

$$E_a = 52.86 \text{ kJ mol}^{-1} //$$